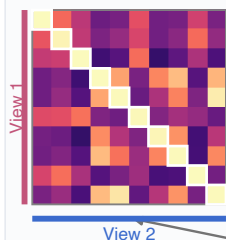


a

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + 0.05 \mathcal{L}_{\text{InfoNCE}}$$

InfoNCE

cross-view similarity matrix
outlined diagonal = matched views



reconstruction of CpGs
withheld from each view

Decoder



CLS representations

View 1

View 2

Shared MethyLLama encoder

6 transformer blocks · 256-d · genomic-rank RoPE

subset 1 · 24,578 CpGs

subset 2 · 24,578 CpGs

two independent
50% CpG subsets

Pretraining corpus

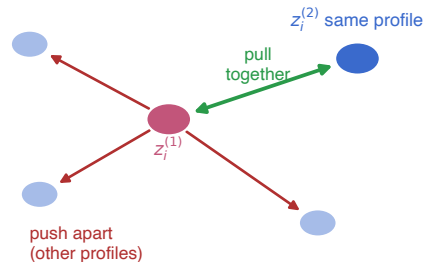
169,120 profiles × 49,156 CpGs

profiles

CpG sites

b

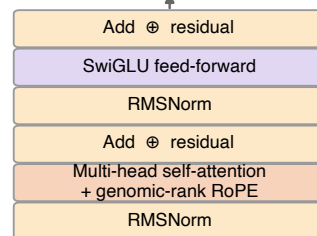
Contrastive objective



c

One transformer block

CLS output = sample representation



d

Token construction

